Response to the reviewer (sixth round)

A reliability-constrained, multi-objective framework for minimum-mass design of steel frame structures under material and loading uncertainty

[Author name]

We thank the reviewer for pointing out that our fifth-round letter did not respond to the comment on marginal versus simultaneous confidence statements. We apologise for the omission; the comment is answered here. The responses to Comments 2–4 in the fifth-round letter are unchanged. Apart from the new simultaneous bounds below, which are computed from the existing verification samples, no numerical results have changed. Section and table numbers refer to the revised manuscript.

Comment 1 (marginal versus simultaneous confidence statements). *The one-sided 95% lower bounds are marginal bounds for individual limit states. Insert the suggested "Scope of confidence statements" paragraph after the description of RBDO-C95, and make the Abstract and Conclusions consistent. The statement that all seven RBDO-C95 designs meet both targets with 95% confidence should be qualified as a set of separate marginal assessments or supported by a simultaneous confidence procedure.*

Response. We did both.

- *Scope paragraph.* The suggested paragraph is inserted at the end of Section 2.5, immediately after the description of RBDO-C95, with the reviewer's wording. We added one sentence: where the paper makes a simultaneous statement, it uses the Bonferroni procedure. Each of m bounds is computed at the one-sided level $1 - 0.05/m$, so that all m hold together with probability of at least 95% whatever the dependence between the estimates. The definition of claim (L) now also says that it is a marginal statement about one limit state of one design.
- *Simultaneous procedure (tool).* We added `beta_lcb_simultaneous` to `rbo/reliability.py` and the table generator `tab_simultaneous` to `make_figures.py`. The latter writes `results/simultaneous.json` and the new Table 16. The bounds use the existing 10^6 verification samples, so no new simulation was needed. The results for the RBDO-C95 designs are:
  - for each design on its own ($m = 3$ limit states), all bounds exceed their targets, so each design meets its three targets simultaneously with at least 95% confidence;
  - for all seven designs together ($m = 21$), all bounds exceed their targets except the ULS bound of the low-wind frame, which is 2.995 against a target of 3.0 (marginal bound 3.005). A simultaneous 95% statement for all seven frames together therefore cannot be made. We report this as it is.
- *Abstract.* It now says that the point-estimate and confidence-bounded designs met *each* target with *marginal* 95% confidence in the baseline surrogate model. It adds that each RBDO-C95 design also met its three targets simultaneously (Bonferroni), but not all seven designs jointly.
- *Results, Discussion and Conclusions.* Every statement that a design meets "both targets with 95% confidence" now reads "each target with marginal 95% confidence". This covers Sections 4.2,

4.4 and 4.6, the caption of Table 13, the Discussion and Conclusions 3–4. Section 4.6 has a new paragraph "Simultaneous confidence" with Table 16. Conclusion 4 states the per-design simultaneous result and the exception for the seven-design family. The statement that the bounds address Monte Carlo sampling uncertainty only, and not model-form or systematic approximation errors, is retained in each place.

The marginal and simultaneous bounds differ by less than 0.03 in β_{ULS} and less than 0.01 in β_{drift}. The distinction therefore changes a conclusion only for designs within a few hundredths of a target, as the low-wind frame shows.

Remaining item

The author and affiliation placeholders remain, because the names and affiliations have not yet been supplied to us. They will be inserted before submission. The manuscript keeps the exact repository commit that produced it.